

Some Radical Viewpoints in Physics

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Abstract

This is a summary of unpopular, widely rejected, even puzzling assumptions. Despite this, they may hint what physics beyond.

Every physical phenomena is described by observer. (Galileo G.)

It means the physical world we describe is not real, but reflects the real geometry behind. A common misreading is “if we don’t see the Moon, the Moon is not there”, it’s fundamental like existence and causality in philosophy.

Space is relation between objects. (G. W. Leibniz)

Unlike Newton’s (and current) view, Leibniz thought space is not absolute. If there is no object, there is no space.

Accelerated positive charge generates gravitational field opposite accelerating direction. (M. Faraday, X. Q. Zhang)

After induction, Faraday tried to find the relation between electromagnetism and gravity. In modern view, it’s neglected since gravity is 10^{-40} weaker than electromagnetism.

Second law of thermodynamics is not strong law. (L. Boltzmann, E. Schrödinger)

With statistical interpretation, entropy is just maximized with largest probability, life tends to reduce it. A common objection is the entropy of total system is still increased, but the entropy of universe is not well-defined.

Mass is equivalent to curvature of spacetime. (W. K. Clifford, A. Einstein)

Despite being successful, major physicists still prefer QFT rather GR except in cosmology (they have no choice!).

Acceleration is not absolute, but relative to observer. (E. Mach)

It was believed by Einstein and included in original GR, but modern texts removed it due to lack of formulation.

Every physical quantity is effect of space motion. (E. Schrödinger, X. Q. Zhang)

Zitterbewegung explains mass, spin, and magnetic moment as effect of spiral motion, but it has never been observed. Before Higgs mechanism, every particle is assumed to be massless, all is contributed from its interaction.

Spin is angular momentum of field, it's field in rotating.

It's actually included in some QFT texts just QM preferred phrasing *intrinsic property*, an illustration is photon has two circular polarizations: left-handed and right-handed.

Electromagnetism plays important role in any scale. (H. Alfvén)

It seemingly explains coronal heating problem and galaxy rotation curve, but contradicts Big Bang. In literature, it's usually referred "universe is a great electricity network, star is like light bulb."

Every object, or its surrounding space does spiral motion outward in lightspeed. (X. Q. Zhang)

Since everything does spiral motion, Zhang suggested it as a postulate to explain every physical phenomena. For example "time is our feeling of such motion", hence the time of light is freezing.

Mass is the number of space moving outward. (X. Q. Zhang)

An illustration is two photons moving in opposite direction give stationary effective mass. In SR, it has $4\pi Gm = - \oint \mathbf{G} \cdot d\mathbf{n}$, analog electric field line.

Charge is the number of space moving outward per time. (X. Q. Zhang)

It's most puzzling of Zhang's claim since $q = \frac{dm}{d\tau}$ clearly violates observation. Maybe it's similar to current, or oscillation, varying in direction.

Mass can be varied via artificial field. (X. Q. Zhang)

Applications include cold-welding and lightspeed traveling. Whether in classical or relativistic, mass is implicitly assumed to be invariant. A common misreading is "we can't accelerate an object to lightspeed", but it's reducing mass to 0.

Magnetic monopole doesn't exist. (X. Q. Zhang)

After Dirac deduced charge quantization, major physicists were trying to search for magnetic monopole. Zhang interpreted pole as face of plane, hence always dipole.

Big Bang doesn't exist. (X. Q. Zhang, Linvo)

Due to existence of singularity, most physicists thought GR isn't complete. However, Big Bang itself relies on it, contradicts such belief. Zhang thought universe is infinitely large, no beginning, no ending.

Hubble's Law is observer effect. (X. Q. Zhang, me)

Zhang thought vacuum/dark energy is due to presence of observer, hence it has minimum $T = \frac{3}{2}\rho c^2$, put into Newton's gravity gives $H_0^2 = 4\pi G\rho$. An illustration is we can't block everything since obstacle itself also interacts us.

Neutrino is the carrier of weak interaction. (J. S. Hei)

It explains the mass of neutrino come from electromagnetism, hence the generation number equals space dimension. Standard Model treats neutrino as massless, and Higgs particle is strange compare to others.

CMB is heat effect of universe expansion. (Espen Gaarder Haug, Stéphane Wojnow)

Using naïve gravity statistics, they derived $kT_H = \frac{\hbar c}{4\pi\sqrt{2l_p}R_H}$ agreeing with observed value. It has never been calculated correctly in history, and it's objected merely due to conflict with Big Bang.

Spacetime isn't quantized. (L. Zhao)

As being successful, some QG assume spacetime is quantized. However, it contradicts Lorentz covariance (rotation doesn't make sense!), subverts QFT itself. Also it has never been observed, with upper limit 10^{-48}m .

Quantized is equivalent to periodic. (me)

Since every quantization of gravity failed, I suggest quantization isn't needed. A quantity is quantized just reflects some periodicity behind, hence spin, charge, color are quantized.

Hubble's constant has running coupling behaviour. (me)

To resolve classical deviation, also Hubble tension, I suspect it has anomaly due to heat motion, analog QFT. As distance shorter, it goes to infinity.

Every conserved quantity (current) corresponds to an interaction. (me)

As the field equation of gravity, electromagnetism, strong interaction relates momentum, charge, color, I suggest there are always such correspondence. Hence angular momentum should be related to an unknown interaction, which may explain galaxy rotation curve.

Reference

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